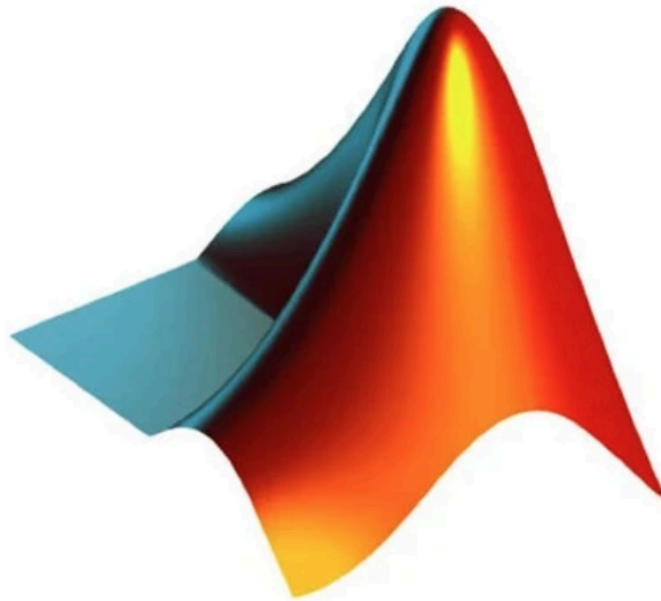
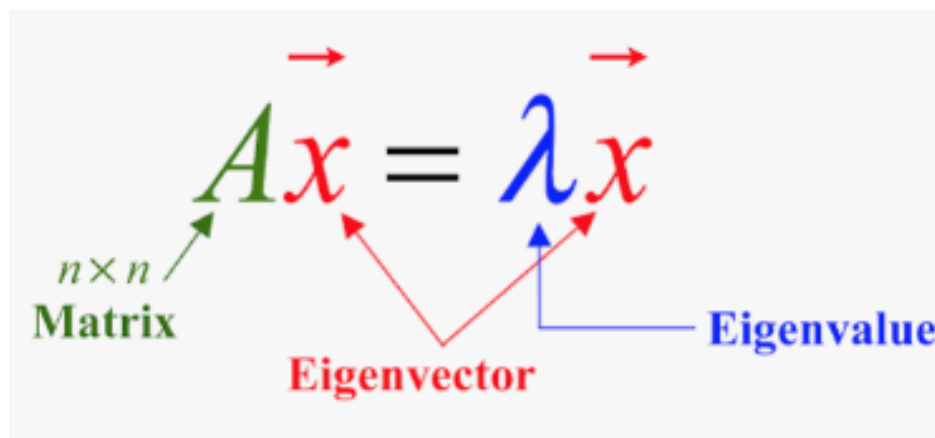


MATH 284 MATLAB Project #4 - Eigenvalue, Eigenvector & Transformation Studio



MATLAB®

$$(\lambda_1, v_1) = \left(1, \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \right) \text{ and } (\lambda_2, v_2) = \left(-1, \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \right)$$

A diagram illustrating the eigenvalue equation $A\vec{x} = \lambda\vec{x}$. The matrix A is green, $\vec{x}$ is red, and λ is blue. Red arrows point from the labels 'Matrix' and 'Eigenvector' to A and $\vec{x}$ respectively. A blue arrow points from the label 'Eigenvalue' to λ . Above each vector $\vec{x}$ is a red arrow indicating its direction.

$A\vec{x} = \lambda\vec{x}$

$n \times n$
Matrix

Eigenvector

Eigenvalue

Project Objective

Develop an interactive MATLAB application that allows users to compute eigenvalues and eigenvectors, visualize invariant directions under transformations, diagonalize matrices when possible, and explore how eigenvalues affect geometric transformations.

The emphasis is to demonstrate why eigenvalues and eigenvectors matter in linear algebra and other fields, not solely simply calculating them.

Concepts And Textbook Chapters/Sections Covered

Chapter 5: Eigenvalues and Eigenvectors

- 5.1 Eigenvectors and Eigenvalues
 - 5.2 The Characteristic Equation
 - 5.3 Diagonalization
 - 5.4 Eigenvectors and Linear Transformations
-
- Eigenvalues
 - Eigenvectors
 - Characteristic Equation
 - Characteristic Polynomial
 - Matrix Diagonalization
 - Similar Matrices
 - Linear Transformations
 - Geometric Interpretation of Eigenvectors

GUI Layout

Eigenvalue & Transformation Studio	

Matrix Input	
Matrix A	
[	]
[Analyze Matrix]	

Eigenvalue Analysis	
Eigenvalues	
Eigenvectors	
Characteristic Polynomial	
Determinant	
Trace	

Diagonalization	
Matrix P	
Matrix D	
P ⁻¹	
Verify A=PDP ⁻¹	

Visualization	
Original Grid	
Eigenvectors	
Transformation Animation	

Tools	
[Characteristic Polynomial]	
[Diagonalize]	
[Transformation Viewer]	
[Animate]	

Necessary MATLAB Files For This Project

EigenStudioApp.m

Responsibilities:

- *Build interface*
- *Coordinate calculations*
- *Display graphics*
- *Handle user interaction*

EigenAnalyzer.m

Stores:

- *the matrix A*
- *eigenvalues*
- *eigenvectors*
- *characteristic polynomial $p(z)$*
- *trace (sum of main diagonal entries)*
- *determinant*

CharacteristicPolynomial.m

Computes:

- *$\det(A - \lambda I) = 0$*

Features:

- *Symbolic polynomial*
- *Roots*
- *Multiplicities*

DiagonalizationAnalyzer.m

Computes:

- *Matrix P (matrix of eigenvectors)*
- *Matrix D (matrix of eigenvalues)*
- *Inverse of P (P^{-1})*

And verifies $A = PDP^{-1}$

EigenTransformationViewer.m

Visualizes:

- *Original vectors*
- *Eigenvectors*
- *Transformed vectors*

EigenVisualizer.m

Responsible for:

- *Coordinate grid*
- *Eigenvector display*

- *Transformation animation*
- *Scaling along eigenvector directions*

Necessary MATLAB Methods For This Project

createComponents()
 analyzeMatrix()
 diagonalizeMatrix()
 displayResults()
 updatePlots()
 clearGUI()
 computeEigenvalues()
 computeEigenvectors()
 computeCharacteristicPolynomial()
 verifyEigenpair()
 buildPolynomial()
 findRoots()
 factorPolynomial()
 isDiagonalizable()
 computeP()
 computeD()
 verifyDiagonalization()
 applyTransformation()
 plotTransformation()
 animateTransformation()
 highlightEigenvectors()

GUI Functions

- Eigenvalue solver
- Characteristic polynomial calculator
- Diagonalization explorer
- Animated transformation viewer

Pseudocode

START PROGRAM

Create GUI

Initialize

Matrix input

Results panel

Plot panel

WHEN Analyze Matrix pressed

Read matrix

Compute

Eigenvalues

Eigenvectors

Characteristic Polynomial

Trace

Determinant

Display results

WHEN Diagonalize pressed

Determine if diagonalizable

IF true

Compute

P

D

P^{-1}

Verify PDP^{-1}

ELSE

Display reason

WHEN Transformation Viewer pressed

Apply matrix

Plot

Original vectors

Eigenvectors

Transformed vectors

Animate transformation

WHEN Clear pressed

Reset GUI

END PROGRAM

0

1

8

9

1

9

7

3

9

8

2

5

1

7

3

8

Row A

Col A

Nul A

5

8

1

0

2

3

5

0

9

8

0

0

6

7

2

4

linear algebra

$\lambda_1 = 3$

$\lambda_2 = 7$

$\det(A) = 0$